Isaac (Zac) Adjei

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[Date]

[Hiring Manager Name/Recruitment Team]
[Company Name]
[Company Address/Location]

Re: Application for [Role Title] at [Company Name]

Dear [Hiring Manager Name/Recruitment Team],

I am writing to apply for the [Role Title] position at [Company Name]. As an Electronic Engineering and Computer Science student at Aston University (Predicted First Class) and Top 40 Finalist in the Black Heritage Undergraduate of the Year Award 2026, I build things end to end, from bare-metal firmware on microcontrollers through to full-stack web applications, CI/CD pipelines and hardware PCB design, and I have shipped every project I have started.

My range of project work demonstrates that breadth. At the software end, my personal portfolio dashboard (Next.js 16, Supabase, Cloudflare Workers) integrates 30+ live data sources: PS5 and Discord real-time presence, Spotify now-playing, WakaTime coding activity, blog analytics and automated CV generation via GitHub Actions. At the hardware end, I built PHAEMOS, an IoT predictive maintenance platform with ESP32, STM32, Arduino Nano and Raspberry Pi Pico 2W hardware nodes running firmware I wrote from scratch; the backend applies Isolation Forest anomaly detection to 18 real-time sensor streams at sub-200ms latency. In between: a 4×4×4 NeoPixel LED cube with adaptive brightness, a two-stage audio amplifier with a custom PCB verified against a 281mW output target, a CNC control system with a hardware E-Stop interrupt and a watchdog timer, and avr-zac (1,200 lines of bare-metal ATmega644P C with no libraries and no OS). I am also the author of git-unlocked, a 217-topic MIT-licensed Git curriculum published as a citable research artefact with a permanent DOI on Zenodo, and my dotfiles project brings consistent tooling (Starship, nvm, accessibility-first colour scheme) across macOS, Windows and Linux. I am drawn to [Company Name] because of [specific reason] and I am excited to contribute to [specific area or challenge].

My Electronic Engineering degree gives me a mathematical foundation that a pure software education would not: signal processing, control systems, power electronics and analogue design taught me to reason about physical constraints, not just logical ones. That crossover perspective, of understanding both the hardware a system runs on and the software that drives it, is genuinely unusual at student level and I have found it makes me a more careful, more complete engineer. I complement technical depth with operational habits: CI/CD pipelines, version-controlled dotfiles, documented workflows and a commitment to writing code that a future reader can understand.

I am looking for a placement or internship where I will be expected to contribute meaningfully from day one, not be given toy tasks. [Company Name] 's [specific team, product area or engineering culture] is the kind of environment where I would grow fastest, and I am the kind of person who leaves systems better than I found them.

Thank you for considering my application. I look forward to the opportunity to discuss how my hardware-to-software breadth can add value to your team.

Yours sincerely,

Isaac (Zac) Adjei